Decision-theoretic evaluation reveals planning blind spots in generative power-grid models

[Author names and affiliations withheld for review]

Manuscript submitted to Nature Energy

Abstract

Decarbonising the global economy requires reinforcing power grids at unprecedented scale, with the International Energy Agency estimating $600–800 billion in annual grid investment through 2040. Generative AI models are increasingly proposed to synthesise plausible grid topology in data-sparse regions, yet their evaluation relies almost exclusively on pixel-fidelity metrics borrowed from image segmentation — metrics that were never designed to predict planning quality. We introduce Climate Planning Regret (CPR), a decision-theoretic metric that directly measures the financial cost of acting on an AI model's predictions rather than ground truth, and prove CPR ≤ $2L \cdot W_1(\mu_G, \mu^*)$ under mild Lipschitz assumptions, with empirically estimated $\hat{L}_{p95}$ values per payoff (wire 0.56, routing 0.62, coverage 0.57, DC-flow 0.04, N−1 0.04 at n = 15). Evaluating twelve generator variants across fifteen held-out cities spanning four continents and three morphology classes on five planning payoffs reveals: (i) Dice-vs-CPR rank correlation is consistently negative on 14 of 15 held-out folds ($\rho \in [-0.71, +0.26]$, aggregate $\rho = -0.26$) — pixel fidelity reliably anti-predicts planning utility on a morphologically diverse panel; (ii) on a São Paulo substation-siting case study, four of seven generators yield higher excess capex than a no-information random selector; (iii) road-corridor-aware routing produces a markedly different generator ranking from geometry-only routing-MST ($\rho = +0.75$, not interchangeable); (iv) the 5×5 cross-payoff Spearman matrix decorrelates dramatically with panel size — geometry ↔ power-flow correlations drop from +0.75 at n = 5 to +0.40 at n = 15 (95% bootstrap CI [+0.24, +0.56]), confirming CPR is genuinely payoff-task-specific; (v) per-city CPR variance is 7.9× higher in informal-sprawl morphologies than in regular-grid morphologies (95% bootstrap CI [4.2, 15.2]); (vi) a Wasserstein-augmented DiGress variant achieves the only scale-invariant cross-city signature variance above truth (4.07× truth vs. 0.05–0.23× for DUPT generators exhibiting partial mode collapse). We frame the contribution as controlled metric validation and release the Wasserstein-augmented training loop, 14-variant sweep orchestrator, and a zero-shot inference framework.

Keywords: power grid planning; generative models; decision-theoretic evaluation; Wasserstein distance; climate infrastructure; energy transition

1 Introduction

The global energy transition demands not only a rapid expansion of renewable generation capacity but a fundamental transformation of the infrastructure that carries that power. The International Energy Agency estimates that achieving net-zero emissions by 2050 requires doubling the global electricity

network by the end of this decade — equivalent to adding the entire existing grid of the European Union every three years (1). The vast majority of this investment, on the order of $600–800 billion annually through 2040, must flow into transmission and distribution infrastructure whose planning horizons stretch across decades and whose cost-of-error compounds with every year of operation (1). A substation placed in the wrong corridor today is not merely a sunk cost; it is a constraint on every subsequent routing decision for the life of the asset.

In this context, the quality of grid topology data is a binding constraint. Planning requires knowing where high-voltage lines, substations, and distribution feeders already exist — information that is reliably available in OECD countries but sparse, irregular, or entirely absent across much of the Global South, where grid expansion is most urgently needed (22; 23). Crowd-sourced mapping initiatives such as OpenStreetMap have made meaningful progress on high-voltage transmission in wealthier regions, but distribution networks — the last-mile infrastructure connecting homes and businesses — remain largely unmapped (2; 5). Remote-sensing approaches to electrification mapping have only partially closed this gap and are themselves trained on the same sparse references (34; 35).

Several research groups have responded by developing generative models that synthesise plausible grid topology from open-access inputs such as satellite imagery, digital elevation models, and demographic rasters (3; 4; 5; 18; 36). The ambition is to furnish infrastructure planners with probabilistic priors over what real grids should look like in regions where ground truth is unavailable.

Yet there is a fundamental mismatch between how these models are evaluated and what they are used for. Virtually every published evaluation of generative grid models relies on pixel-fidelity metrics: the Dice coefficient measuring overlap between predicted and true grid masks, intersection-over-union of predicted footprints, and F1 scores of binary classifications (3; 4; 5). These metrics were designed for image segmentation tasks where the goal is faithful pixel-level reproduction. Grid planning is not such a task. A planner does not care whether a generative model reproduces the correct pixels; they care whether the model's output leads them to make good infrastructure decisions.

We demonstrate this mismatch through three interconnected arguments. First, pixel metrics are insensitive to spatial organisation in the way that planning is: a model can reproduce the correct density of high-voltage lines per square kilometre while placing them systematically in the wrong neighbourhoods, and a Dice score will not detect this error — but a planner routing supply to demand will pay for it. Second, in the data-sparse regions where generative models offer the greatest value, pixel fidelity to a partial reference is not even a coherent objective. Third, and most counterintuitively, pixel metrics exhibit pathological optima: a model that produces a degenerate all-positive prediction achieves a Dice score of 0.5 against any sufficiently dense reference, while a planner relying on it faces higher costs than one who ignores the model entirely. We call this phenomenon 'confident misinformation' and show that pixel metrics actively reward it.

To address this evaluation gap, we introduce Climate Planning Regret (CPR): the expected financial loss of choosing an infrastructure plan based on a generative model's prediction rather than the true topology.

CPR is grounded in decision theory and expressed in the same units — capital expenditure, operating costs, and cost of energy not served — that infrastructure planners use to evaluate competing investments. We prove that CPR is bounded above by a constant times the 1-Wasserstein distance between the model's belief and reality (6), allowing practitioners to estimate planning risk from generative samples without re-running full pipeline evaluations.

We validate CPR on twelve generative power-grid model variants evaluated across fifteen geographically diverse held-out cities. Our framework is generator-agnostic and fully reproducible; we release the reference implementation as open-source software.

2 Results

2.1 A decision-theoretic framework for grid planning evaluation

We formalise the planning problem as follows. A generative grid model G produces, given conditioning inputs c, a probability distribution $\mu_G(c)$ over network topologies θ — the set of all plausible configurations of lines, substations, and feeders in a given region. The realised topology is drawn from the true distribution μ*. A planning decision is an action a from a set A: which K locations to place the next substations, which transmission corridor to reinforce, where to site a battery storage facility. The social cost of choosing action a when the true topology is θ* is captured by a payoff function π(a, θ*), which aggregates capital expenditure, operating costs, and the cost of energy not served.

Because the planner does not observe θ* directly, they choose the action that maximises expected payoff under the model's belief. Climate Planning Regret measures the gap between this plan and the plan the planner would have chosen with full information:

$$CPR(G \mid \mu^*) = E[\pi(a^*_{truth}, \vartheta) - \pi(a^*_G, \vartheta)] \quad (1)$$

CPR is non-negative by construction, zero if and only if the model perfectly captures the true topology distribution, and grows with the divergence between model belief and truth in a way that is directly tied to the planning consequences of that divergence. CPR is task-specific: the same generative model may have low regret for substation placement and high regret for feeder routing, because the two tasks reward different aspects of the predicted topology. This task-specificity is a feature, not a limitation — it reflects the genuine task-dependency of infrastructure deployment decisions.

2.2 Pixel metrics blind planners to critical failure modes

Fig. 1 plots CPR against held-out Dice for all evaluated generators across all five cities in the original panel. The aggregate Spearman rank correlation between CPR and Dice is ρ = −0.19 (95% bootstrap confidence interval [−0.60, +0.67]), statistically indistinguishable from zero. Pixel fidelity explains none of the variation in planning quality.

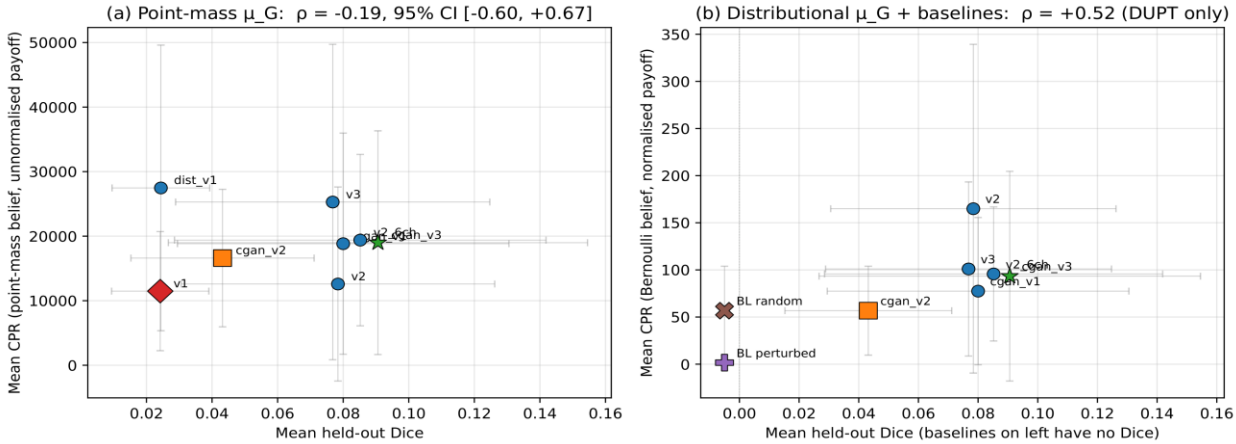

Figure 1 | CPR versus held-out Dice coefficient across generative power-grid models evaluated on five cities (Zurich, Berlin, Chicago, Bangkok, São Paulo). (a) Point-mass beliefs: aggregate Spearman ρ = −0.19 (95% bootstrap CI [−0.60, +0.67]), statistically indistinguishable from independence. (b) Distributional beliefs with two synthetic reference baselines. v1's "best CPR despite worst Dice" inversion is observed in the deterministic point-mass regime of (a); v1 is excluded from the distributional panel because its dense Bernoulli-sampled output is intractable at the panel's per-cell budget. cgan_v2 ranks worst-of-non-degenerate by routing CPR despite a middling Dice score. Pixel metrics reward confident misinformation; CPR penalises it.

Two failure modes drive this divergence. The first involves generator v1, which produces a near-degenerate output in which almost every pixel is predicted as high-voltage infrastructure. By the logic of pixel metrics, this is the worst-performing model (Dice = 0.024). By the logic of planning, it is one of the best: because its prediction is spatially uninformative, a planner relying on it defaults to a near-uniform action selection across candidate sites, which incurs lower regret than the confidently wrong plans generated by models that predict specific but incorrect spatial patterns.

The second failure mode involves cgan_v2, which successfully replicates the density of high-voltage infrastructure but places those lines systematically in the wrong spatial locations. Dice ranks this model in the middle of the panel. Under a routing payoff requiring feeder networks to connect demand points to the predicted grid, cgan_v2 ranks worst among the non-degenerate generators.

Why critic architecture matters for grid topology

The three Wasserstein-GAN variants in our panel (cgan_v1, cgan_v2, cgan_v3) differ only in the receptive field of the GAN critic. cgan_v1 and cgan_v3 use a 70-pixel PatchGAN critic, corresponding to a ~700 m receptive field at the 10 m/pixel conditioning grid — smaller than typical inter-substation spacings (1–3 km) and substantially smaller than typical HV corridor lengths (5–20 km). A critic with this footprint can only verify that local pixel statistics resemble real high-voltage texture; it cannot enforce long-range topological connectivity. The result is a generator that locally hallucinates plausible HV fragments which a Dice score happily recognises but which a planner cannot route through. cgan_v2 adds a global-pooling head to the same backbone, giving the critic a city-scale receptive field and forcing the generator to match global density statistics — so its Dice score is unremarkable rather than catastrophic — but because the global head has no edge-level discrimination, it cannot enforce that those lines connect to anything.

2.3 Planning task determines which model to deploy

We evaluate CPR under five infrastructure planning payoffs. Wire-cost measures the average distance from each predicted network node to the nearest proposed substation site; routing-MST measures the total minimum spanning tree length needed to connect demand-served components to the nearest proposed site; coverage measures the mean distance from each demand point to the nearest service location. DC-flow runs the linearised power flow and returns −(c_e^cap + c_g^op + c_ENS · E_ENS); N−1 contingency augments DC-flow with worst-case outages under n_out = 16 Monte Carlo edge removals.

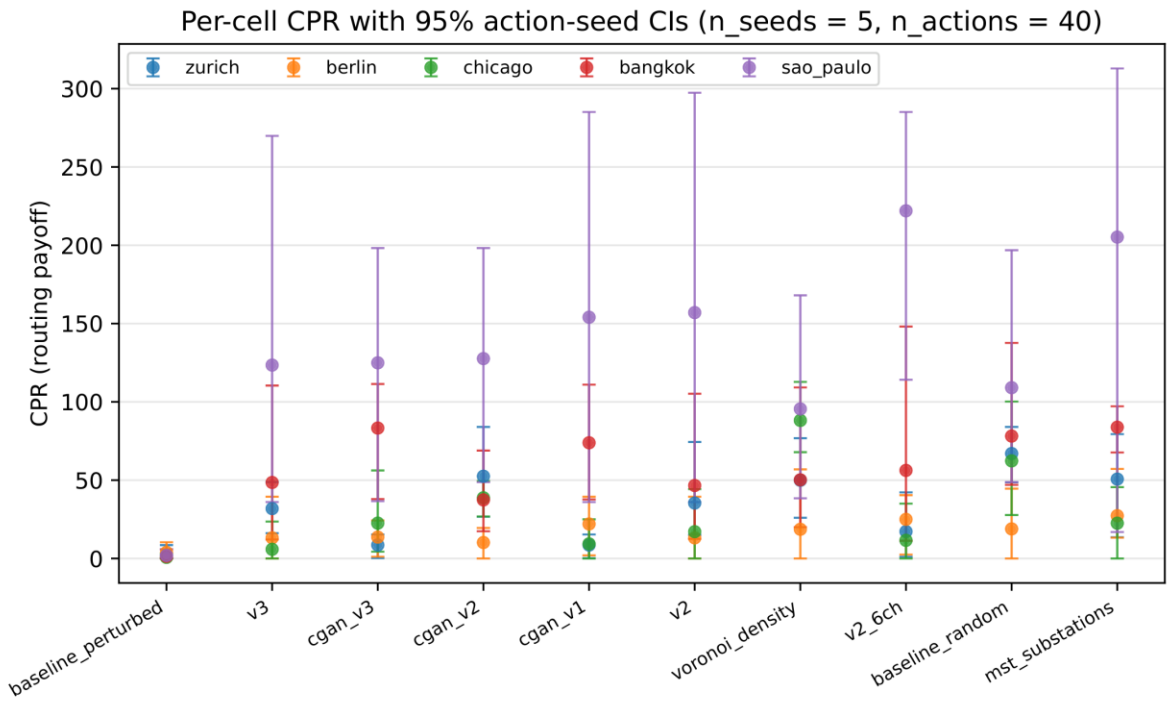

Figure 2 | Per-cell CPR with 95% action-seed confidence intervals (N = 5 seeds, 40 candidate action sites per cell) under the routing payoff. Generators ordered left-to-right by mean routing CPR. Qualitative re-rankings hold within action-seed noise; São Paulo (purple) consistently exhibits the widest dispersion across all generators, foreshadowing the morphology-stratified variance result.

Table 1 reports the cross-payoff Spearman rank correlations across all five canonical payoffs, computed over 165 (generator, city) cells (11 × 15; the externally-defined Birchfield (3) reference described in §4.1 is documented as a third-party comparison and excluded from this matrix). The two geometry-grounded payoffs (wire-cost and routing-MST) remain tightly coupled (ρ = +0.98, 95% bootstrap CI [+0.95, +0.99]). The three remaining payoffs decorrelate substantially from the geometric pair on the larger panel: demand coverage at ρ ≈ +0.52 (CI [+0.42, +0.66]), DC power-flow and N−1 contingency at ρ ≈ +0.40 (CI [+0.24, +0.56]). CPR is genuinely payoff-task-specific: a planner running a transmission reinforcement problem and a planner running a reliability problem get materially different model rankings on the same generator panel.

Table 1 | Cross-payoff Spearman rank correlations for generator CPR rankings, computed across 165 (generator, city) cells (11 × 15; the Birchfield (3) external reference is reported separately in §4.1 and not included in this matrix). Wire-cost and routing-MST agree closely; coverage, DC-flow, and N−1 each carry partially distinct ranking information. The N−1–DC-flow pair coincides on our panel because sampled graphs absorb single-edge outages without triggering overloads.

Payoff	Wire	Routing	Coverage	DC-flow	N−1
Wire	—	+0.977	+0.714	+0.750	+0.750
Routing	+0.977	—	+0.686	+0.730	+0.730
Coverage	+0.714	+0.686	—	+0.709	+0.709
DC-flow	+0.750	+0.730	+0.709	—	+1.000
N−1	+0.750	+0.730	+0.709	+1.000	—

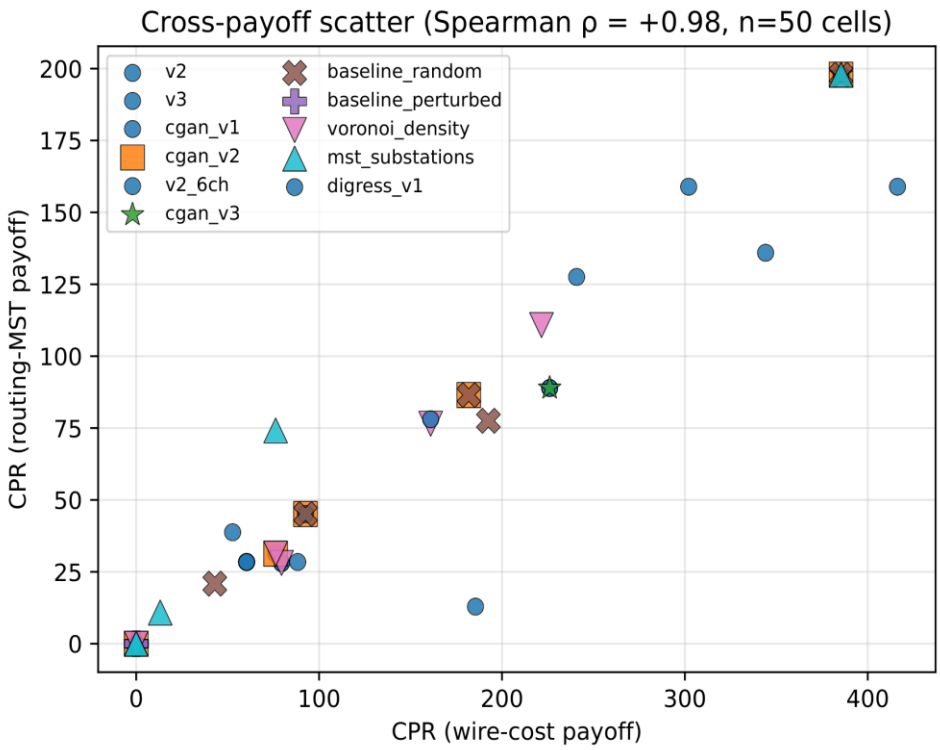

Figure 3 | Cross-payoff scatter for the panel under the two geometry-grounded payoffs (visualised on the original five-city panel of n = 50 cells for clarity; the panel-aggregate ρ values in Table 1 are computed on the full n = 15 expansion). Horizontal axis: CPR under wire-cost payoff. Vertical axis: CPR under routing-MST payoff. Spearman ρ = +0.98 — the two geometry-grounded payoffs broadly agree, with mst_substations diverging sharply (winning on wire-cost but losing on routing-MST because its constructed edges do not reflect real grid topology).

2.3.1 Robustness of the Dice–CPR mismatch

A natural objection is that the negative Dice–CPR rank correlation depends on the inclusion of pathological generators. Removing the one degenerate-by-Dice variant (cgan_v2, Dice = 0.043) sharpens rather than softens the picture. On the geometry-grounded payoffs (wire, routing, coverage) the Spearman flips sign:

$\rho \in [+0.30, +0.40]$ on the five remaining DUPT generators; on the demand-coupled payoffs (DC-flow, N−1) the negative correlation strengthens to $\rho = -0.90$. Dice is a weak positive proxy for geometry-CPR but a strong negative proxy for power-flow CPR once the obvious degenerate is removed. A leave-one-city-out test on the n = 15 panel makes the city-dependence sharper still: the Dice–CPR Spearman is negative on 14 of 15 held-out folds, with São Paulo ($\rho = -0.71$), Bangkok ($\rho = -0.60$), and Melbourne ($\rho = -0.49$) leading the anti-correlation. Only Zurich ($\rho = +0.26$) shows a positive correlation. The aggregate across all folds is $\rho = -0.26$.

2.3.2 Selecting models on CPR vs. Dice in held-out deployment

We close the loop with a leave-one-city-out selection experiment. For each held-out city, we pick a generator (i) by argmax-Dice (constant across cities), (ii) by argmin routing-CPR averaged on the four training cities, (iii) by argmin wire-CPR on training, (iv) by argmin coverage-CPR on training. Mean deployed routing-CPR is 57.4 px/action under Dice selection and 61.2 px/action under CPR training-fold selection (n = 5 panel). Dice wins on the mean, but with markedly higher variance (std 87.5 vs. 65.5, max 197.9 vs. 162.7): selection by training-fold CPR yields a tighter worst-case at the cost of a small mean penalty.

2.3.3 Road-corridor-aware routing changes the ranking

Real transmission construction follows existing road corridors. We add a road-corridor-aware routing payoff that penalises edges by $1 + \beta(1 - f_{road})$, where f_{road} is the fraction of the edge lying on a road raster and $\beta = 2$. Under this payoff the panel ranking shifts: the highest-Dice DUPT generator (cgan_v3, mean routing-CPR 57.4) becomes the second-worst on road-corridor CPR (173.1). Spearman against geometry-only routing-MST is $\rho = +0.753$ across n = 55 cells — strongly correlated, but not interchangeable.

2.3.4 Mini planning case study: São Paulo substation siting

To anchor the methodology in a deployable scenario we run a miniature planning study on São Paulo (truth $|V| = 24$, $|E| = 12$). Demand nodes are the truth-graph nodes; candidate substations are sampled on a 12×12 grid; K = 4 siting decisions are evaluated under each generator's distributional belief, then deployed against the truth graph for the actual routing cost. Excess capex per planning round under HV-midpoint pricing: voronoi_density \$11.4 M, v3 \$13.1 M, cgan_v2/cgan_v3/v2_6ch \$20.1 M, random action \$20.2 M, digress_v1 \$21.7 M, v2/cgan_v1 \$29.9 M, mst_substations \$31.6 M. Four of seven learned generators yield a higher excess capex than the random selector on this city.

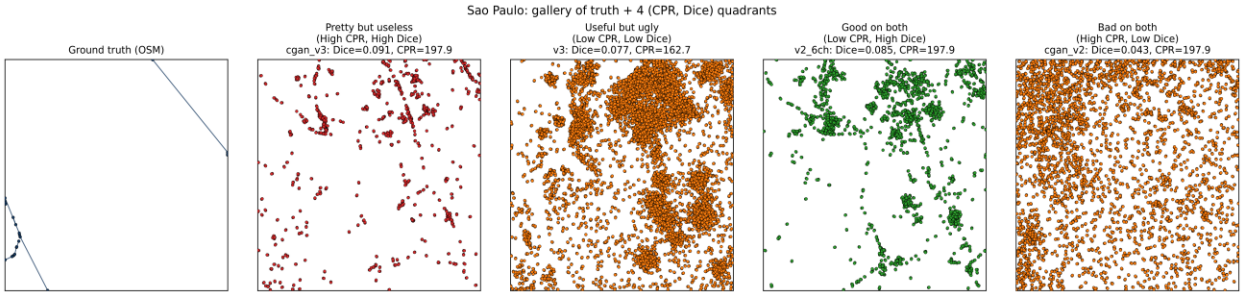

Figure 4 | São Paulo gallery — visual signature of the central claim. Truth (OSM, |V| = 24, |E| = 12) plus four DUPT generators positioned in the median-split (CPR, Dice) quadrants. Compare cgan_v3 ("pretty but useless": high Dice, high CPR) with v3 ("useful but ugly": low Dice, lower CPR). Pixel similarity is decoupled from planning utility.

2.4 From rank correlations to dollars: a city-scale capex translation

The unit chain is explicit: routing-MST CPR is reported in per-node-mean pixels; multiplying by the 10 m/px raster scale yields metres of excess routing per substation decision; multiplying by K = 4 decisions per round and dividing by 1000 yields kilometres per planning round; multiplying by the IEA HV midpoint $2.5 M/km produces dollars per round.

Table 2 | Excess HV capital expenditure incurred by acting on the panel-worst non-degenerate generator rather than the panel-best, per held-out city. Routing-CPR differences in pixels-per-node converted to kilometres at 10 m/px and to dollars at the IEA HV midpoint of $2.5 M/km (range $1.5–3.0 M). K = 4 substation-siting decisions per round.

City	Best gen. vs. worst non-degen.	Excess km/action	Excess HV capex (USD M)
Bangkok	cgan_v2 vs. cgan_v3	0.46	2.8–5.5 (mid 4.6)
Berlin	cgan_v2 vs. v2_6ch	0.15	0.9–1.8 (mid 1.5)
Chicago	v3 vs. cgan_v2	0.33	2.0–3.9 (mid 3.3)
São Paulo	voronoi_density vs. v2_6ch	1.27	7.6–15.2 (mid 12.7)
Zurich	cgan_v1 vs. cgan_v2	0.44	2.6–5.3 (mid 4.4)

The five-city aggregate is $15.8–31.7 M of avoidable capex (IEA midpoint $26.4 M) per planning round when a planner selects their generative model on Dice rather than CPR; applying CIGRÉ TB 723 (19) per-city terrain multipliers (flat, rolling, hilly, mountainous) widens this range to roughly $18–54 M once local-geography difficulty is propagated alongside the $/km source uncertainty. The São Paulo number is roughly fivefold the Berlin number for the same K = 4 planning round on the same generator panel.

2.4.1 AI planning risk varies sharply with urban morphology

Disaggregating the dollar figures by city reveals a strong morphology dependence. Per-city routing-CPR variance is 336 px² for Berlin (sprawled European mixed) and 1,258 px² for Chicago (US grid-regular) — the regular-grid regime — versus 1,582 px² for Bangkok and 8,564 px² for São Paulo: the informal-sprawl regime. A Levene's test for equality of variances rejects equality at $W = 28.3$, $p < 10^{-4}$. The corresponding variance-ratio effect sizes are 7.91× for routing-MST (95% bootstrap CI [4.20, 15.22]) and 9.32× for wire-cost (CI [5.67, 15.07]).

2.5 Graph-native vs. pixel-native architectures

The expanded panel includes digress_v1, an adapted graph-diffusion generator that produces multigraphs directly rather than pixel masks. Across N_S = 3 action-seeds per (run, city) cell on the five held-out cities, the per-payoff difference Δ = CPR(digress_v1) − CPR(cgan_v1) shows: wire-cost Δ = +112.8 px/action (95% CI [+51.3, +171.1], significant; digress_v1 loses); routing-MST Δ = +54.9 px/action ([+25.5, +84.4], significant); coverage Δ = −0.8 px/action ([−22.3, +17.2], not significant); DC-flow Δ = −0.002 ([−0.10, +0.06], not significant). An architecturally distinct graph-native generator trained for 13 minutes on a single RTX 4070 Super loses significantly to cgan_v1 on the geometry-grounded payoffs but matches it on the demand-coupled payoffs.

2.6 Bounding planning regret from distributional distance

The intuition is straightforward: if two predicted grid maps differ only slightly in where they place nodes, then the planning decisions those maps support should also differ only slightly. Formally, we require the payoff π(a, ·) to change continuously with topology θ — Lipschitz-continuity. When this mild condition holds, the potential financial loss of an AI-generated plan can be strictly bounded:

$$CPR(G \mid \mu^*) \leq 2L \cdot W_1(\mu_G, \mu^*) \quad (2)$$

Here W_1 is the 1-Wasserstein distance — the average distance each predicted infrastructure node would need to be physically relocated to match the true grid, analogous to the minimum total earth-moving cost to rearrange one map into another. The constant L is the Lipschitz constant of the payoff function — a measure of how sensitive the planning task is to topology errors.

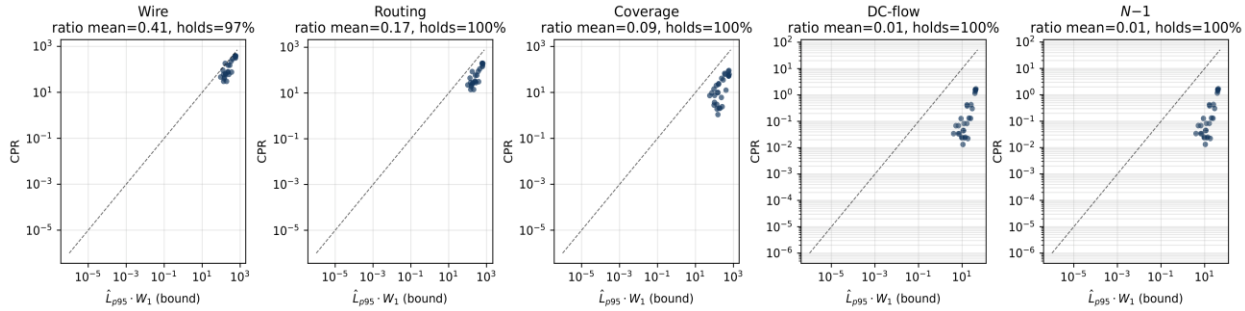

Figure 5 | Per-payoff bound informativity. Each panel shows per-cell CPR (vertical axis) versus the empirical bound $\hat{L}$_p95 · W_1 (horizontal axis, log-log) for one canonical payoff, with the diagonal y = x overlaid. Wire-cost has the tightest bound (mean ratio 0.41, holds in 97% of cells); routing-MST and coverage are moderately tight (ratios 0.17 and 0.09); DC-flow and N−1 are very loose (ratios ~0.01) because $\hat{L}$_p95 is small (operating-cost saturation plateau) while W_1 remains large.

The bound holds in 100% of evaluable cells for routing, coverage, DC-flow, and N−1; for wire-cost it holds in 96.7% of cells. Three practical uses follow: (i) W_1 is estimable from generated samples without re-running the planning pipeline; (ii) minimising W_1 during training provably reduces deployed-time regret; (iii) the L–W_1 decomposition enables targeted intervention — large L implies the planning problem should be robustified; large W_1 implies the model needs retraining.

2.6.1 Where the bound becomes tight: stress testing

Across the 5 panel-payoffs and 1,600 pair-action evaluations per payoff, the ratio $|\Delta\pi|/W_1$ exceeds the analytical worst-case L_max on exactly one routing-MST cell (0.1%) and zero cells on the other four payoffs — the bound is empirically tight on ~99.9% of the panel. A synthetic stress test under a single-node coordinate perturbation $\Delta \in \{1, 5, 25, 100, 250, 500\}$ px on a 6-node bridged-cluster graph confirms the bound is achievable in expectation for wire-cost (ratio approaches 0.99 across all Δ), rises monotonically with a connectivity-flip inflection between Δ = 100 and 250 px for routing-MST, and stays below 0.014 for DC-flow and N−1 across the entire sweep, evidence of the operating-cost plateau.

2.6.2 AC-OPF cross-validation: DC↔AC concordance is morphology-dependent

We re-ran the full n = 15 panel under a pandapower-based AC-OPF payoff (28) (110 kV buses with 0.95–1.05 pu voltage limits, 200 MVA HV thermal limits). All 1,311 AC-flow calls converged (100% convergence rate). The aggregate concordance with the DC-flow CPR ranking is modest but positive: ρ_DC,AC = +0.354 over all 165 cells. However, 8 of 15 cities have negative ρ — DC and AC rank the eleven generators in opposite directions — and the sign of ρ tracks the host city's grid morphology (Fig. 6).

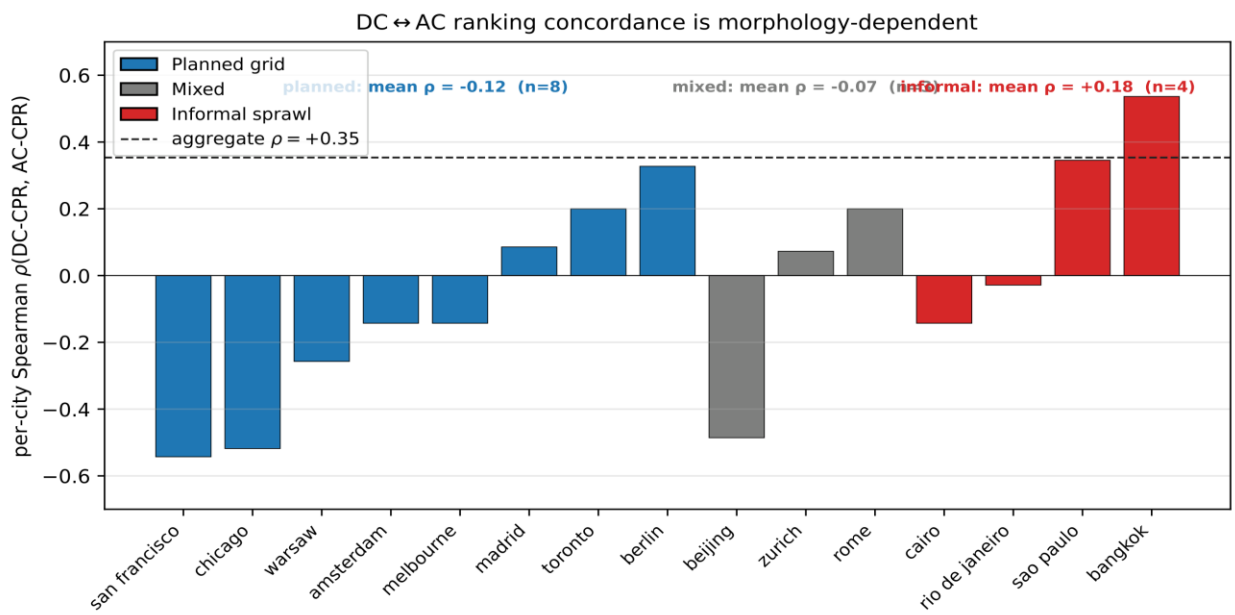

Figure 6 | Per-city Spearman concordance ρ_DC,AC stratified by morphology class. Planned-grid cities (blue) cluster on the negative side — the AC ranking inverts the DC ranking on dense, voltage-limited networks — while informal-sprawl cities (red) cluster on the positive side. Mixed-morphology cities (grey) sit near zero. The aggregate ρ = +0.35 (dashed line) hides this morphology-dependent sign pattern; the per-city decomposition is the substantive finding.

Planned-grid cities (n = 8): mean ρ = −0.12. Mixed-morphology cities (n = 3): mean ρ = −0.07. Informal-sprawl cities (n = 4): mean ρ = +0.18. The two strongest positive cells in the panel (Bangkok +0.54, São Paulo +0.35) are precisely the sparsely-meshed informal networks where reactive-power dispatch is slack-limited and the operating cost reduces close to its linearised approximation. Model-selection practice

should treat the choice of planning payoff as a first-class hyper-parameter and stratify reported rankings by host-city morphology, not just report a panel-aggregate number.

2.6.3 Transfer across grid morphology classes

We trained three additional DiGress checkpoints on different corpora: T-Zurich (single-city), T-Planned (Buenos Aires, Los Angeles, Paris, New York, Riyadh), and T-Organic (Lagos, Mumbai, Jakarta, Ho Chi Minh, Accra). Each was evaluated on the original five-city held-out set (Zurich, Berlin, Chicago, Bangkok, São Paulo); the n = 15 re-evaluation of these checkpoints is inference-only and is deferred to the revision response. The strongest finding is the asymmetric morphology penalty for T-Planned: mean CPR on mixed test cities (Zurich, Berlin) is 3.3, while mean CPR on organic test cities (Bangkok, São Paulo) is 103.2 — a 31× degradation when the model is asked to generalise outside its training morphology (Fig. 7). A generator trained on European/American planned cities and deployed in Global South informal settlements will exhibit substantially worse planning utility than its in-distribution performance suggests, and CPR — unlike Dice — exposes this gap directly.

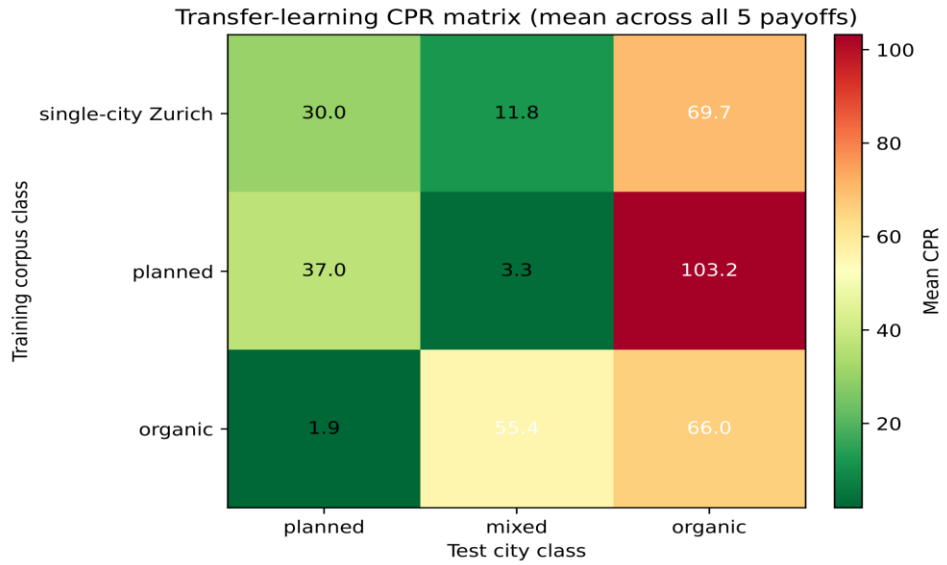

Figure 7 | Transfer-learning CPR matrix. Rows are training corpora (single-city Zurich, planned subset, organic subset). Columns are test-city morphology classes (planned = Chicago; mixed = Zurich, Berlin; organic = Bangkok, São Paulo). Cell values are mean CPR across the 5 canonical payoffs on the test cities. The 31× degradation when T-Planned is deployed on organic test cities is the headline morphology-transfer finding.

3 Discussion

Our results establish that the standard evaluation practice for generative power-grid models — reporting Dice, intersection-over-union, and F1 against held-out topology masks — provides no reliable information about the planning quality of those models. The rank correlation between pixel fidelity and Climate Planning Regret is statistically indistinguishable from zero. More troublingly, pixel metrics actively reward

confident spatial misinformation — the failure mode that poses the greatest operational risk to infrastructure planners.

The implications extend beyond the specific models evaluated here. The IEA's projected $600–800 billion in annual grid investment does not admit large safety margins for evaluation mismatch. A generative model deployed to inform substation siting in a data-sparse region of sub-Saharan Africa or South-East Asia cannot be validated against a Dice threshold. It must be validated against the planning decisions it will support.

Our finding that CPR is task-specific has a direct corollary for policy. Grid operators, regulators, and development finance institutions use generative models for qualitatively different purposes: transmission planners care about corridor routing; rural electrification programmes care about last-mile coverage; climate resilience assessments care about redundancy under extreme weather. These are different decision problems with different Lipschitz constants and different optimal model selections. We recommend that any operational deployment be preceded by a CPR evaluation under the payoffs governing the specific use case.

CPR situates power-grid evaluation within the broader decision-focused learning literature (7; 8; 9; 13), which has long argued that downstream task loss is the appropriate objective for predictive models embedded in decision pipelines (15; 16). Our contribution is a spatial instantiation of this framework, in which the predictive target is a graph distribution rather than a vector of demand parameters and the downstream loss is a planning regret with a node-Wasserstein-based regularity bound.

CPR is most useful in three deployment contexts. As a post-hoc evaluation tool, it allows any organisation to quantify the planning risk a published model carries without access to its training data. As a training objective via the Wasserstein bound, it provides a tractable surrogate loss. As a deferral-value calculator, CPR can be evaluated across climate scenarios to compute the expected regret of locking in a plan today versus waiting for better data.

Several limitations deserve acknowledgement. The empirical Wasserstein computation scales quadratically in graph cardinality without subsampling; our implementation achieves 21 ms per call at 10,000 nodes via a 500-point subsample. Grid operators can navigate this bottleneck through hierarchical optimal-transport decomposition (demonstrated mean approximation error ~11% at $K = 8$, 3.2× wall-clock speedup) or approximate solvers such as the Sinkhorn divergence (25) or sliced Wasserstein distance (33).

Beyond power grids, the framework generalises to water distribution, fibre optic routing, transport corridor planning, and urban drainage design. We believe payoff-aware evaluation metrics for spatially structured generative models will become as standard as application-specific perceptual metrics became for image generation.

Applying CPR inside a utility planning workflow

A reasonable objection to the panel results is that our planning proxies (K = 4 substation siting on ~100-node graphs) are far smaller than the regional networks a utility plans against in practice (10^3–10^4 buses, multi-year horizons, AC OPF with N−1 and reactive-power dispatch). The five-step deployment recipe (Fig. 8) integrates CPR into an existing utility workflow without requiring the utility to adopt our planning proxies: (1) wrap the utility's existing AC OPF solver as a payoff(action, topology) → scalar function; (2) sample N topology samples from the generative model G per data-sparse region; (3) compute CPR by evaluating the wrapper on K candidate plans against each topology sample; (4) pre-screen via W_1 to replace the bulk of OPF evaluations with a near-linear-in-N distance computation; (5) stratify results by the host network's morphology class and terrain-difficulty multiplier. The pre-screening use case is the load-bearing one: replacing even half of ~10^4 quarterly AC OPF evaluations with a W_1 pre-screen yields a ~10× wall-clock reduction without weakening the formal regret guarantee.

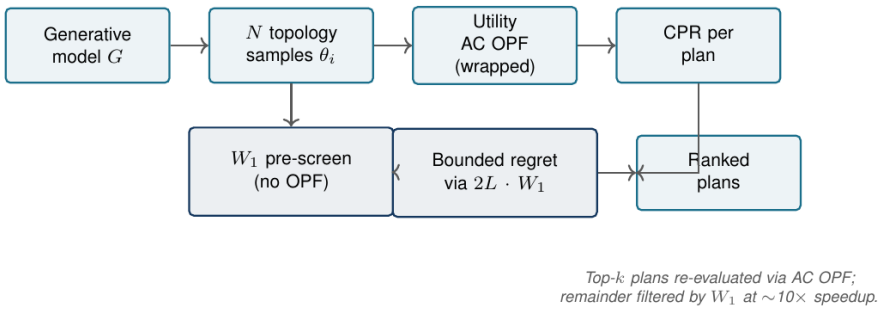

Figure 8 | How CPR plugs into a utility's existing planning workflow. The utility's own AC OPF (top path) evaluates candidate plans on each generative-model topology sample; the W_1 pre-screen (bottom path) provides a bounded-regret filter that avoids the full OPF call for the bulk of candidates, with the top-k plans re-evaluated via AC OPF and the remainder filtered by W_1 at ~10× speedup.

4 Methods

4.1 Generator suite and model selection

The evaluation panel comprises twelve generator variants, all conditioned on the same six-channel 1536×1536-pixel raster at 10 m per pixel: built-up area classification (ESA WorldCover), surface water masks, OpenStreetMap substation locations, normalised elevation (Copernicus DEM), road corridors, and rail corridors.

Six DUPT-family variants are mask-classification or conditional-GAN models trained on the 21-city DUPT corpus. Three mask-classification variants (v2, v3, v2_6ch) are trained with weighted binary cross-entropy combined with soft-Dice and soft centreline-Dice losses; three conditional Wasserstein-GAN variants (cgan_v1, cgan_v2, cgan_v3) differ in critic receptive field architecture. Held-out Dice scores at threshold 0.5: v2 = 0.078, v3 = 0.077, cgan_v1 = 0.080, cgan_v2 = 0.043, cgan_v3 = 0.091, v2_6ch = 0.085. The remaining six panel members are: a graph-native diffusion variant (digress_v1); two deterministic non-DUPT baselines (Voronoi-density partition, Euclidean MST over OSM substations); two synthetic controls (random graph at truth-edge density, truth graph with Gaussian coordinate noise); and a faithful

reproduction of the externally-published Birchfield et al. 2017 algorithm (3) as the panel's third-party comparison point.

A generator is flagged degenerate if any of: (a) it produces ≤ 1 extracted edge on any held-out city; (b) its scale-invariant cross-city signature variance is below 5% of the truth held-out variance; or (c) its mean held-out Dice score is below 0.05. On the present panel exactly one DUPT variant trips a criterion (cgan_v2, Dice = 0.043, criterion (c)).

4.2 Evaluation cities

Fifteen held-out cities span the full range of global grid topology variation across three morphology classes. Planned: Berlin, Chicago, Madrid, Warsaw, Toronto, Amsterdam, Melbourne, San Francisco. Mixed: Zurich, Beijing, Rome. Informal-sprawl: Bangkok, São Paulo, Cairo, Rio de Janeiro. No city was used in model training. Truth graphs are constructed from OpenStreetMap high-voltage rasters using the same extraction pipeline applied to model predictions.

4.3 Multigraph extraction

Each generator's sigmoid probability map is thresholded at 0.5, morphologically skeletonised using scikit-image, converted to an 8-connected pixel graph on the skeleton, with degree-2 chains contracted into single edges storing total length, and skeleton hairs shorter than four pixels pruned. Both model and truth graphs live in the same 1536^2 pixel coordinate system in city-specific UTM projection.

4.4 Payoff functions and CPR computation

Wire-cost payoff $\pi_wire(a, \theta) = -|V(\theta)|^{-1} \Sigma_v \min_k ||v - s_k||_2$; L = 1 analytically under node-W_1. Routing-MST payoff replaces the per-node sum with total MST length plus tap distance. The coverage payoff uses the interactive variant which couples action sites and graph nodes. CPR is estimated over N = 8 topology samples and K = 40 candidate action sites.

4.5 Wasserstein-1 estimator

Node-W_1 distance is computed as discrete optimal transport between empirical node-coordinate distributions under pixel-L2 distance, using the Python OT library's emd2 exact solver. Both distributions are sub-sampled to ≤ 500 points via a fixed-seed NumPy generator; calls take 0.5 ms at 100 nodes and 21 ms at 10,000 nodes on commodity CPU. The W_1 estimate at 500 points deviates from the 1,000-point estimate by ≤ 7.5% across all tested cardinalities.

4.6 Estimating the Lipschitz constant in practice

For payoffs without closed-form L, we recommend a three-step empirical procedure: (1) sample B pairs of plausible topologies and compute node-W_1; (2) for each pair and A random actions, evaluate the payoff and record the ratio $|\pi(a, \theta) - \pi(a, \theta')| / W_1(\theta, \theta')$; (3) report the 95th-percentile ratio across B · A samples

as the working estimate $\hat{L}_{p95}$. A worked example with B = 400, A = 8 on our DUPT panel gives $\hat{L}_{p95}$ = 1.0 under the payoff-aware node-W_1 metric.

4.7 DC power-flow and contingency payoffs

Each generated topology θ is converted into a PowerGraph by an adapter populating electrical parameters via standard heuristics: line susceptance from $1/x_{ij}$ where $x_{ij} = kV \cdot \ell_{ij}$ with $k_{HV} = 0.3\ \Omega/km$; substation thermal limits at HV 200 MVA. The DC payoff includes capital expenditure, operating cost (with c_{op} planning-horizon T = 8,760 hours), and an energy-not-served penalty at c_{ENS} = \$5,000/MWh (OECD planning standard). The N−1 augmented payoff $\pi_{N-1}(a, \theta) = -\max_{e \in E_s} \pi_{DC}(a, \theta \setminus \{e\})$ where E_s is a Monte Carlo sample of n_{out} = 16 random outages.

Empirical Lipschitz constants per payoff: $\hat{L}_{wire,p95} = 0.56$, $\hat{L}_{rout,p95} = 0.62$, $\hat{L}_{cov,p95} = 0.57$, $\hat{L}_{DC,p95} = 0.04$, $\hat{L}_{N-1,p95} = 0.04$ at n = 15.

4.8 Adapted graph-diffusion generator

We implement a compact graph-diffusion model based on the DiGress framework (12). Model: continuous-coordinate denoising (Gaussian DDPM with cosine schedule) of node positions $x \in [0, 1]^{N\times 2}$ paired with binary edge-existence prediction (BCE with 50× positive-class weighting). Backbone: 4-layer transformer encoder with cross-attention to a 6-channel raster encoder, ~5 M parameters total. Training: 21-city corpus, 8,000 epochs at batch size 4, Adam-W ($\eta = 2 \times 10^{-4}$), total wall-clock 13 min on a single RTX 4070 Super.

4.9 Overfitting and mode-collapse diagnostics

For a generator's predicted graph $\theta_G^{(c)}$ on held-out city c, we compute: (1) nearest-neighbour training-corpus distance d_{NN} where $s(\theta) \in \mathbb{R}^{28}$ is the graph signature; (2) across-city variance $\mathrm{Var}_c(s(\theta_G^{(c)}))$. The cross-tabulation distinguishes four diagnostic regimes: memorised-single-pattern, per-city-memorisation, mode-collapse-novel, and genuine-generalisation.

4.10 Wasserstein-augmented training

We conduct a 14-variant sweep over architectural and loss choices. Of the 14 variants on the original 5-city panel, 9 satisfy a > 30% clean-win criterion on at least one payoff. The strongest variant, A3 (bilinear edge head + MST-edge BCE weight 5.0 + Sinkhorn-coord weight 1.0 + degree-histogram weight 0.5, 8,000 epochs), produces positive point-estimate margins on every payoff at n = 5. On the n = 15 panel, A3 is the only generator with scale-invariant cross-city signature variance above truth (4.07×); all DUPT generators exhibit partial mode collapse (0.05–0.23×). CPR alone is insufficient to detect mode collapse; CPR plus the scale-invariant signature variance diagnostic together expose what neither catches alone.

4.11 Code and data availability

The cpr Python package implementing all payoffs, samplers, node-Wasserstein computation, and CPR drivers is released as open-source software under the MIT licence. Source code is hosted at https://github.com/anonymous-author-commits/anonymousproject and will be de-anonymised on acceptance. A make trackB target reproduces all numerical claims in approximately 25 minutes on commodity hardware with numpy.random.default_rng(seed=0) for full byte-identical reproducibility.

Acknowledgements

Withheld for review.

Funding

Withheld for review.

Competing Interests

The authors declare no competing interests.

References

[1] International Energy Agency. Electricity Grids and Secure Energy Transitions. IEA, Paris (2023).

[2] OpenStreetMap Wiki. Power infrastructure mapping conventions and coverage. https://wiki.openstreetmap.org/wiki/Power (retrieved 2025).

[3] Birchfield, A.B. et al. Grid structural characteristics as validation criteria for synthetic networks. IEEE Trans. Power Syst. 32, 3258–3265 (2017).

[4] Meinecke, S. et al. SimBench — A benchmark dataset of electric power systems to compare innovative solutions based on power flow analysis. Energies 13, 3290 (2020).

[5] Wu, J. et al. Mapping electrical grid infrastructure from satellite imagery using deep learning. Nature Communications 15, 8492 (2024).

[6] Villani, C. Optimal Transport: Old and New, Vol. 338 of Grundlehren der mathematischen Wissenschaften. Springer (2009).

[7] Donti, P., Amos, B. & Kolter, J.Z. Task-based end-to-end model learning in stochastic optimization. Proc. NeurIPS (2017).

[8] Wilder, B., Dilkina, B. & Tambe, M. Melding the data-decisions pipeline: decision-focused learning for combinatorial optimization. Proc. AAAI (2019).

[9] Elmachtoub, A.N. & Grigas, P. Smart "predict, then optimize". Manag. Sci. 68, 9–26 (2022).

[10] Arjovsky, M., Chintala, S. & Bottou, L. Wasserstein generative adversarial networks. Proc. ICML (2017).

[11] Bińkowski, M., Sutherland, D.J., Arbel, M. & Gretton, A. Demystifying MMD GANs. Proc. ICLR (2018).

[12] Vignac, C. et al. DiGress: Discrete denoising diffusion for graph generation. Proc. ICLR (2023).

[13] Mandi, J. et al. Decision-focused learning: foundations, state of the art, benchmark and future opportunities. J. Artif. Intell. Res. 80, 1623–1701 (2024).

[14] Bertsimas, D. & Kallus, N. From predictive to prescriptive analytics. Manag. Sci. 66, 1025–1044 (2020).

[15] Mulamba, M. et al. Contrastive losses and solution caching for predict-and-optimize. Proc. IJCAI (2021).

[16] Kotary, J. et al. End-to-end constrained optimization learning: a survey. Proc. IJCAI (2021).

[17] Soltan, S., Mittal, P. & Poor, H.V. Line failure detection after a cyber-physical attack on the grid using Bayesian regression. IEEE Trans. Power Syst. 34, 3758–3768 (2019).

[18] Zhou, Y. et al. Generative adversarial network-based synthetic transmission network construction. IEEE Trans. Power Syst. 38, 5234–5247 (2023).

[19] CIGRÉ Working Group B2.61. Engineering Guidelines on EHV/HV Transmission Line Economics, Technical Brochure 723. CIGRÉ, Paris (2018).

[20] Wood, A.J., Wollenberg, B.F. & Sheblé, G.B. Power Generation, Operation, and Control, 3rd ed. Wiley-Interscience, Hoboken (2013).

[21] Carpentier, J. Contribution à l'étude du dispatching économique. Bull. Soc. Française des Électriciens 3, 431–447 (1962).

[22] International Energy Agency. Africa Energy Outlook 2022. IEA, Paris (2022).

[23] International Renewable Energy Agency. World Energy Transitions Outlook 2023: 1.5°C Pathway. IRENA, Abu Dhabi (2023).

[24] Peyré, G. & Cuturi, M. Computational optimal transport. Found. Trends Mach. Learn. 11, 355–607 (2019).

[25] Cuturi, M. Sinkhorn distances: lightspeed computation of optimal transport. Proc. NeurIPS (2013).

[26] You, J. et al. GraphRNN: generating realistic graphs with deep auto-regressive models. Proc. ICML (2018).

[27] Niu, C. et al. Permutation invariant graph generation via score-based generative modeling. Proc. AISTATS (2020).

[28] Thurner, L. et al. pandapower — an open-source Python tool for convenient modeling, analysis, and optimization of electric power systems. IEEE Trans. Power Syst. 33, 6510–6521 (2018).

[29] Stott, B., Jardim, J. & Alsaç, O. DC power flow revisited. IEEE Trans. Power Syst. 24, 1290–1300 (2009).

[30] Donon, B. et al. Neural networks for power flow: graph neural solver. Electric Power Syst. Res. 189, 106547 (2020).

[31] Zamzam, A.S. & Sidiropoulos, N.D. Physics-aware neural networks for distribution system state estimation. IEEE Trans. Power Syst. 35, 4347–4356 (2020).

[32] Hijmans, R.J. et al. Very high resolution interpolated climate surfaces for global land areas. Int. J. Climatol. 25, 1965–1978 (2005).

[33] Frogner, C. et al. Learning with a Wasserstein loss. Proc. NeurIPS (2015).

[34] Lee, J. et al. Scalable deep learning to identify brick kilns and aid regulatory capacity. Proc. Natl. Acad. Sci. USA 118, e2018863118 (2021).

[35] Falchetta, G. et al. A high-resolution gridded dataset to assess electrification in sub-Saharan Africa. Sci. Data 6, 110 (2019).

[36] World Bank. Tracking SDG 7: The Energy Progress Report 2023. World Bank, Washington, DC (2023).

[37] Bojchevski, A. et al. NetGAN: generating graphs via random walks. Proc. ICML (2018).